Structured notes: Module 2, electromagnetic induction and Maxwell's equations

Griffiths, *Introduction to Electrodynamics* (5th ed.), Chapter 7. Figures are redrawn for these notes, not copied from the book; the book figure number is in each caption. Problems are not included.

Section 1: electromotive force (Griffiths 7.1)

7.1.1 Ohm's law

Source: book-sources/m2-s71-ohm.html, Griffiths pp. 298–303. Equation numbers as printed in the book.

Prerequisites

- Current density $\mathbf{J}$; potential difference $V = -\int \mathbf{E} \cdot d\mathbf{l}$.
- Gauss's law; the divergence of a curl is zero; Laplace's equation.

PLUS TWO RECALL: OHM'S LAW AND RESISTIVITY

Recall. From Class 12: $V = IR$, and for a wire $R = \rho\ell/A$, where ρ is the resistivity.

Re-derive.

1. Take a uniform cylinder of length ℓ and area A, with potential difference V between its ends. The field is $E = V/\ell$ (uniform, Example 7.3).
2. The current density is $J = I/A$.
3. From $V = IR$ and $R = \rho\ell/A$: $I = VA/(\rho\ell)$.
4. Therefore $J = I/A = V/(\rho\ell) = E/\rho = \sigma E$, with $\sigma = 1/\rho$.

Saved. $\mathbf{J} = \sigma\mathbf{E}$ (Eq. 7.3).

1. Constitutive law

Let $\mathbf{f}$ be the force per unit charge that drives the charges. For most substances, $\mathbf{J}$ is proportional to $\mathbf{f}$:

$$\mathbf{J} = \sigma\mathbf{f} \qquad (7.1)$$

σ is the conductivity; $\rho = 1/\sigma$ is the resistivity (Table 7.1). Metals: $\sigma = \infty$. Insulators: $\sigma \approx 0$.

For an electromagnetic drive, $\mathbf{f} = \mathbf{E} + \mathbf{v} \times \mathbf{B}$, so $\mathbf{J} = \sigma(\mathbf{E} + \mathbf{v} \times \mathbf{B})$ (Eq. 7.2). Neglecting the magnetic term:

RESULT 1.1: OHM'S LAW

$$\mathbf{J} = \sigma\mathbf{E} \qquad (7.3)$$

In a perfect conductor, $\mathbf{E} = \mathbf{J}/\sigma = \mathbf{0}$ even when a current flows, so connecting wires are equipotentials.

2. Resistance of a uniform cylinder (Ex. 7.1, Fig. 7.1)

For a cylinder of cross-sectional area A and length L, with $\mathbf{E}$ uniform (Ex. 7.3), $\mathbf{J} = \sigma\mathbf{E}$ is uniform and

$$I = JA = \sigma EA = \frac{\sigma A}{L}V$$

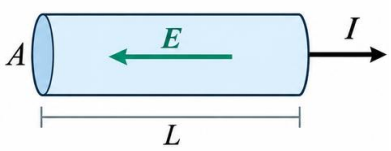


Fig. 7.1 (book): cylindrical resistor, field E along the axis, current I leaving the right end.

3. Resistance of coaxial cylinders (Ex. 7.2, Fig. 7.2)

Field between the cylinders:

$$\mathbf{E} = \frac{\lambda}{2\pi\epsilon_0 s}\hat{\mathbf{s}}$$

where λ is the charge per unit length of the inner cylinder. Then

- $I = \sigma \oint \mathbf{E}\cdot d\mathbf{a} = \frac{\sigma}{\epsilon_0}\lambda L$;
- $V = -\int_b^a \mathbf{E}\cdot d\mathbf{l} = \frac{\lambda}{2\pi\epsilon_0}\ln\frac{b}{a}$.

Hence $I = \frac{2\pi\sigma L}{\ln(b/a)}V$.

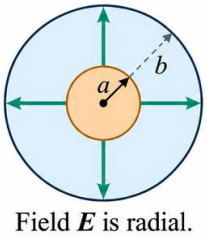


Fig. 7.2 (book): coaxial cylinders, end view; the field E between them is radial.

4. Circuit form

In both cases $I \propto V$:

> **RESULT 1.2: CIRCUIT FORM OF OHM'S LAW**
>
> $$V = IR \tag{7.4}$$
>
> R depends on geometry and σ. Ex. 7.1: $R = L/\sigma A$. Ex. 7.2: $R = \frac{\ln(b/a)}{2\pi\sigma L}$. Unit: Ω (1 Ω = 1 V/A).

5. Proof that the field in Ex. 7.1 is uniform (Ex. 7.3)

V obeys Laplace's equation inside the cylinder. Boundary conditions: $V = 0$ at $z = 0$; $V = V_0$ at $z = L$; $\mathbf{J}\cdot\hat{\mathbf{n}} = 0$ on the curved surface, so $\partial V/\partial n = 0$ there. The function $V = V_0 z/L$ satisfies all of these. By the uniqueness theorem it is the solution. $\mathbf{E} = -\nabla V = -(V_0/L)\hat{\mathbf{z}}$ is uniform.

6. Steady current and charge density (Eq. 7.5)

$$\nabla\cdot\mathbf{E} = \frac{1}{\sigma}\nabla\cdot\mathbf{J} = 0 \tag{7.5}$$

Charge density vanishes inside a homogeneous ohmic conductor carrying a steady current; unbalanced charge resides on the surface. Laplace's equation holds in such a medium.

> **PLUS TWO RECALL: JOULE HEATING**
>
> **Recall.** From Class 12: power $P = VI$, and with $V = IR$, $P = I^2R$.
>
> **Re-derive.**
>
> 1. Work per unit charge across a resistor is V.
> 2. Charge passing per unit time is I.
> 3. So energy per unit time is VI. Substituting $V = IR$ gives I^2R.
>
> **Saved.** $P = VI = I^2R$ (Eq. 7.7).

7. Joule heating (Eq. 7.7)

> **RESULT 1.3: JOULE HEATING LAW**
>
> $$P = VI = I^2R \tag{7.7}$$
>
> P in watts when I is in amperes and R in ohms.

8. Microscopic model (background, Eq. 7.6)

$$\begin{aligned}\mathbf{J} &= nfq\,\mathbf{v}_{\text{ave}} \\ &= \left(\frac{nf\lambda q^2}{2mv_{\text{thermal}}}\right)\mathbf{E}\end{aligned} \tag{7.6}$$

Classical (Drude) model with n molecules per unit volume, f free electrons per molecule, charge q, mass m, mean free path λ. Qualitative only; footnote 2 notes it is not the quantum theory.

7.1.2 Electromotive force

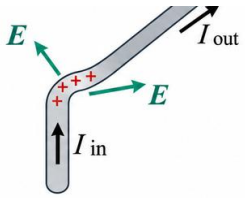
Source: book-sources/m2-s72-emf.html, Griffiths pp. 304–305.

Prerequisites

- Ohm's law and $V = IR$ (Section 7.1.1).
- For an electrostatic field, $\oint \mathbf{E} \cdot d\mathbf{l} = 0$.

1. Current continuity in a circuit

In a steady circuit the current is the same at all points of the loop. An imbalance produces charge accumulation, whose electric field opposes the imbalance (Figs. 7.7, 7.8). The process is quick enough that the current is effectively uniform even at radio frequencies.



Charge at the knee; E points away from it.

Fig. 7.8 (book): charge piles up at a bend, and E points away from the knee.

2. Force decomposition

The total force per unit charge splits into a source part and an electrostatic part:

$$\mathbf{f} = \mathbf{f}_s + \mathbf{E} \tag{7.8}$$

$\mathbf{f}_s$ is non-zero only in the source region. $\mathbf{E}$ acts throughout the circuit.

> **PLUS TWO RECALL: EMF AS WORK PER UNIT CHARGE**
>
> **Recall.** From Class 12: the emf of a source is the work done per unit charge by the source in carrying charge round the circuit.
>
> **Re-derive.**
>
> 1. The source force per unit charge is $\mathbf{f}_s$. Its work per unit charge over one trip round the loop is $\oint \mathbf{f}_s \cdot d\mathbf{l}$.
> 2. The electrostatic field does no net work round a closed loop, since $\oint \mathbf{E} \cdot d\mathbf{l} = 0$.
> 3. So only the source force remains.
>
> **Saved.** $\mathcal{E} = \oint \mathbf{f} \cdot d\mathbf{l} = \oint \mathbf{f}_s \cdot d\mathbf{l}$ (Eq. 7.9).

3. Definition of emf

> **RESULT 1.4: ELECTROMOTIVE FORCE**
>
> $$\mathcal{E} \equiv \oint \mathbf{f} \cdot d\mathbf{l} = \oint \mathbf{f}_s \cdot d\mathbf{l} \tag{7.9}$$
>
> Equality holds because $\oint \mathbf{E} \cdot d\mathbf{l} = 0$ for electrostatic fields. Emf has the unit of volts; it is an integral of force per unit charge, not a force.

4. Ideal source

For an ideal source ($\sigma = \infty$), $\mathbf{E} = -\mathbf{f}_s$ within the source. The terminal potential difference is

$$V = -\int_a^b \mathbf{E} \cdot d\mathbf{l} = \int_a^b \mathbf{f}_s \cdot d\mathbf{l} = \oint \mathbf{f}_s \cdot d\mathbf{l} = \mathcal{E} \tag{7.10}$$

The integral extends over the loop because $\mathbf{f}_s = \mathbf{0}$ outside the source. A real source has internal resistance r and terminal voltage $\mathcal{E} - Ir$ (footnote 4).

> **PLUS TWO RECALL: INTERNAL RESISTANCE**
>
> **Recall.** From Class 12: a real cell of emf $\mathcal{E}$ and internal resistance r gives terminal voltage $V = \mathcal{E} - Ir$ when current I flows.
>
> **Re-derive.**
>
> 1. An ideal source (no resistance) gives terminal voltage $V = \mathcal{E}$ (Eq. 7.10).
> 2. In a real source the current I flows through the internal resistance r, so there is a potential drop Ir inside the source (Ohm's law across r).
> 3. The terminal voltage is the emf less this drop.
>
> **Saved.** $V_{\text{terminal}} = \mathcal{E} - Ir$ (footnote 4).

5. Hydraulic analogy (footnote 5)

Gravity corresponds to $\mathbf{E}$, the pump to the source, and height to voltage.

7.1.3 Motional emf and the flux rule

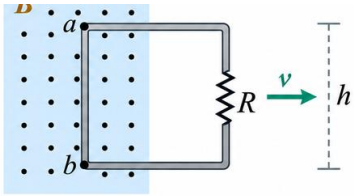
Source: book-sources/m2-s73-motional.html, Griffiths pp. 306–315. Problems not included.

Prerequisites

- Lorentz force $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$; magnetic flux $\Phi = \int \mathbf{B} \cdot d\mathbf{a}$.
- Scalar triple product; Stokes' theorem; divergence.

1. Motional emf in a rectangular loop

Setup (Fig. 7.10): loop side ab of length h lies in a uniform field $\mathbf{B}$ into the page; the loop moves to the right at speed v.



Loop moves right; side ab is in the field region.

Fig. 7.10 (book): rectangular loop with one side in a uniform field B; pulled to the right at speed v.

The magnetic force per unit charge on the moving side is $\mathbf{f}_{\text{mag}} = \mathbf{v} \times \mathbf{B}$, with vertical component vB. Only side ab contributes.

RESULT 5: MOTIONAL EMF

$$\mathcal{E} = \oint \mathbf{f}_{\text{mag}} \cdot d\mathbf{l} = vBh \qquad (7.11)$$

The integral is evaluated at one instant (a snapshot of the loop). Sides bc and ad contribute zero.

PLUS TWO RECALL: MOTIONAL EMF $\mathcal{E} = Blv$

Recall. From Class 12: a conductor of length l moving at speed v perpendicular to a field B has emf Blv.

Re-derive.

1. The magnetic force on a charge q is $q\mathbf{v} \times \mathbf{B}$. So the force per unit charge is $\mathbf{v} \times \mathbf{B}$, with magnitude vB, directed along the wire.
2. Integrate this along the moving side of length h: $\oint \mathbf{f}_{\text{mag}} \cdot d\mathbf{l} = vBh$.

Saved. $\mathcal{E} = vBh$ (Eq. 7.11).

2. Energy balance

The magnetic force does no work on a charge. Let $\mathbf{u}$ be the drift velocity along the wire and $\mathbf{w} = \mathbf{v} + \mathbf{u}$ the resultant velocity. The magnetic force $q\mathbf{w} \times \mathbf{B}$ has a component quB opposing the drift. An external agent exerts $f_{\text{pull}} = uB$ to sustain the current (Fig. 7.11). The work per unit charge along the actual path, of length $h/\cos\theta$, is

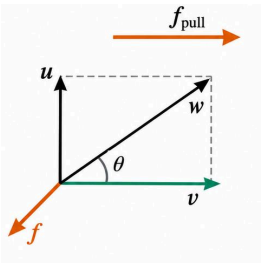


Fig. 7.11 (book): velocity and force vectors for a charge in the moving wire.

$$\begin{aligned} \int \mathbf{f}_{\text{pull}} \cdot d\mathbf{l} &= (uB)\left(\frac{h}{\cos\theta}\right)\sin\theta \\ &= vBh \\ &= \mathcal{E} \end{aligned}$$

The emf is obtained along a different path from the work (Fig. 7.12). The pulling force does the work but contributes nothing to the emf; the magnetic force contributes to the emf but does no work.

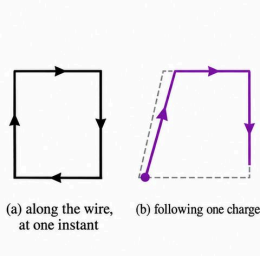


Fig. 7.12 (book): (a) path for the emf, at one instant; (b) path followed by one charge.

3. Flux rule

$$\Phi \equiv \int \mathbf{B} \cdot d\mathbf{a} \tag{7.12}$$

For the loop of Fig. 7.10, $\Phi = Bhx$ and $d\Phi/dt = -Bhv$ (since $dx/dt < 0$). The emf equals minus the rate of change of flux:

RESULT 6: FLUX RULE

$$\mathcal{E} = -\frac{d\Phi}{dt} \tag{7.13}$$

Valid for arbitrary loop shape, motion and field non-uniformity.

PLUS TWO RECALL: FARADAY'S LAW: $\mathcal{E} = -d\Phi/dt$

Recall. From Class 12: the induced emf equals minus the rate of change of magnetic flux, $\mathcal{E} = -d\Phi/dt$.

Re-derive.

1. For the loop of Fig. 7.10, the flux is $\Phi = Bhx$, where x is the length of the loop inside the field.
2. As the loop moves right, x decreases at speed v, so $dx/dt = -v$.
3. Therefore $d\Phi/dt = -Bhv$, and $-d\Phi/dt = vBh$, which is the motional emf of Eq. 7.11.

Saved. $\mathcal{E} = -d\Phi/dt$ (Eq. 7.13).

4. Proof

Let $\mathcal{S}$ and $\mathcal{S}'$ be surfaces bounded by the loop at t and $t + dt$. The change in flux is the flux through the ribbon joining the two loops. A point P on the loop moves to P'; the ribbon element is $d\mathbf{a} = (\mathbf{v} \times d\mathbf{l})\, dt$. Hence

$$\frac{d\Phi}{dt} = \oint \mathbf{B} \cdot (\mathbf{v} \times d\mathbf{l}) = \oint \mathbf{B} \cdot (\mathbf{w} \times d\mathbf{l})$$
$$= -\oint (\mathbf{w} \times \mathbf{B}) \cdot d\mathbf{l} = -\oint \mathbf{f}_{\text{mag}} \cdot d\mathbf{l} = -\mathcal{E}$$

The second equality uses $\mathbf{u} \parallel d\mathbf{l}$; the third uses the scalar triple product.

Sign convention. The direction of $d\mathbf{l}$ and $d\mathbf{a}$ are linked by the right-hand rule. A negative emf indicates current flowing in the negative sense.

5. Limitation: flux-rule paradox (Fig. 7.14)

The flux rule requires a single current path. Throwing the switch from a to b doubles the flux through the circuit but produces no motional emf; the ammeter reads zero. Switches, sliding contacts and extended conductors violate the single-path assumption.

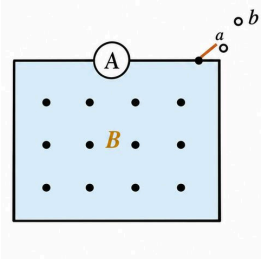


Fig. 7.14 (book): flux-rule paradox; the switch moves from a to b.

6. Example 7.4: Faraday disk

Disk of radius a with angular velocity ω in a field $\mathbf{B}$ along $\hat{\mathbf{z}}$ (Fig. 7.15).

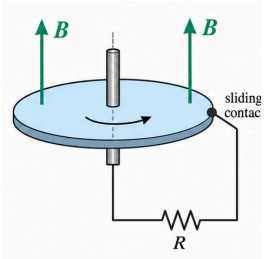


Fig. 7.15 (book): Faraday disk spinning in a field B pointing up.

$$\mathcal{E} = \int_0^a \omega s B \, ds = \frac{\omega B a^2}{2}$$
$$I = \frac{\omega B a^2}{2R}$$

The flux rule is not applicable in its single-path form, because the current spreads across the disk.

7. Eddy currents: Example 7.5

Spherical shell of conductivity σ, radii R_1, R_2, angular velocity $\boldsymbol{\omega}$, uniform $\mathbf{B}$. Current law: $\mathbf{J} = \sigma(\mathbf{E} + \mathbf{v} \times \mathbf{B})$ (Eq. 7.2). Taking the divergence and using Gauss's law and continuity, in the steady state:

$$\rho = -\epsilon_0 \nabla \cdot (\mathbf{v} \times \mathbf{B}) \tag{7.14}$$

With $\mathbf{B} = B\hat{\mathbf{z}}$, $\boldsymbol{\omega} = \omega\hat{\mathbf{z}}$ and the BAC–CAB rule:

$$\mathbf{v} \times \mathbf{B} = \omega B(x\hat{\mathbf{x}} + y\hat{\mathbf{y}}) \tag{7.15}$$

Hence $\rho = -2\epsilon_0\omega B$ (Eq. 7.16) and $\nabla^2 V = 2\omega B$ in the shell (Eq. 7.17), with compensating surface charge at the boundaries.

CASE (A): $\boldsymbol{\omega} \parallel \mathbf{B}$

Radial current vanishes at R_1, R_2 (Eq. 7.18). Solution $V = \frac{\omega B}{2}(x^2 + y^2) + K$ (Eq. 7.21), so $\mathbf{E} = -\mathbf{v} \times \mathbf{B}$ (Eq. 7.22) and $\mathbf{J} = \mathbf{0}$.

$$\begin{aligned} E_r &= -\omega B(x\hat{\mathbf{x}} + y\hat{\mathbf{y}}) \cdot \hat{\mathbf{r}} \\ &= -\omega B r \sin^2\theta \end{aligned} \tag{7.19}$$

$$\begin{aligned} \left.\frac{\partial V}{\partial r}\right|_{R_1} &= \omega B R_1 \sin^2\theta \\ \left.\frac{\partial V}{\partial r}\right|_{R_2} &= \omega B R_2 \sin^2\theta \end{aligned} \tag{7.20}$$

$$V(r,\theta) = \frac{\omega B}{2} r^2 \sin^2\theta + K \tag{7.21}$$

CASE (B): $\boldsymbol{\omega} \perp \mathbf{B}$

$\boldsymbol{\omega} = \omega\hat{\mathbf{x}}$, $\mathbf{B} = B\hat{\mathbf{y}}$: $\rho = 0$, $\nabla^2 V = 0$ in the shell, with $\partial V/\partial r$ specified at R_1 and R_2. Solution $V = -\frac{\omega B}{2} xy + K$. The eddy current $\mathbf{J} = \sigma(\mathbf{E} + \mathbf{v} \times \mathbf{B})$ is non-zero.

Footnote 10: the convection current is smaller than the conduction current by a factor of order $\epsilon_0\omega/\sigma$.

Section 2: electromagnetic induction (Griffiths 7.2)

7.2.1 Faraday's law

Source: book-sources/m2-sec2-faraday.html, Griffiths pp. 316–319. Equations 7.23–7.26; Examples 7.6 and 7.7 (Lenz's law) in full.

Prerequisites

- Magnetic flux $\Phi = \int \mathbf{B} \cdot d\mathbf{a}$ and the flux rule $\mathcal{E} = -d\Phi/dt$ (Eq. 7.13, Section 7.1.3).
- Stokes' theorem: $\oint \mathbf{E} \cdot d\mathbf{l} = \int (\nabla \times \mathbf{E}) \cdot d\mathbf{a}$.
- Lorentz force $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$ (Eq. 7.50).

1. Three experiments

Faraday's experiments of 1831 each produced a current in a loop of wire. Their common result is stated in Section 3.

- **Experiment 1.** The loop is pulled to the right through a magnetic field (Fig. 7.21a). The charges in the moving part of the loop feel the force $q\mathbf{v} \times \mathbf{B}$. This is motional emf (Section 7.1.3); the flux rule gives it directly.
- **Experiment 2.** The magnet is moved to the left and the loop is held still (Fig. 7.21b). The same current flows. Only the relative motion matters, which is why the emf is the same as in Experiment 1. But the loop is at rest. A stationary charge feels no magnetic force, $q\mathbf{v} \times \mathbf{B} = \mathbf{0}$ when $\mathbf{v} = \mathbf{0}$. So the force on the charges must be electric.
- **Experiment 3.** Loop and magnet both at rest, while the field strength is changed (an electromagnet with a varying coil current, Fig. 7.21c). The current flows again. The force is again electric.

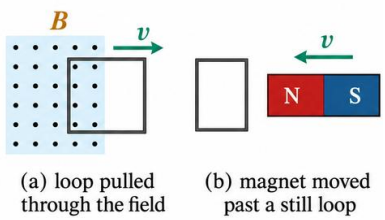


Fig. 7.21 (book): (a) loop pulled through a field; (b) magnet moved past a still loop.

2. The induced electric field

Faraday's inference: **a changing magnetic field induces an electric field.** In Experiments 2 and 3 that induced field drives the charges round the loop. In Experiment 1 the magnetic force does the driving. The book uses the name "Faraday's law" only for the induced electric field.

> **PLUS TWO RECALL: LENZ'S LAW**
>
> **Recall.** From Class 12: the induced current opposes the change in flux that produces it.
>
> **Re-derive.**
>
> 1. Suppose the flux through a loop increases, so $d\Phi/dt > 0$. By Eq. 7.26, $\mathcal{E} = -d\Phi/dt < 0$: the induced emf drives current in the sense that produces a field opposing the increase.
> 2. Suppose the flux decreases, so $d\Phi/dt < 0$. Then $\mathcal{E} > 0$ and the induced field points along the original field, opposing the decrease.
> 3. In both cases the induced current tries to keep the flux unchanged.
>
> **Saved.** The induced current opposes the change in flux (direction only, from Eq. 7.26).

3. Faraday's law in integral and differential form

Assume the emf round a loop equals the rate of change of flux, as Experiments 2 and 3 show. Then

$$\oint \mathbf{E} \cdot d\mathbf{l} = -\frac{d\Phi}{dt} \tag{7.23}$$

Write the flux as a surface integral. Since $\Phi = \int \mathbf{B} \cdot d\mathbf{a}$ and the loop is fixed while $\mathbf{B}$ changes, the time derivative acts on $\mathbf{B}$:

$$\oint \mathbf{E} \cdot d\mathbf{l} = -\int \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{a} \tag{7.24}$$

This is Faraday's law in integral form. Apply Stokes' theorem to the left side: $\oint \mathbf{E} \cdot d\mathbf{l} = \int (\nabla \times \mathbf{E}) \cdot d\mathbf{a}$. Both sides are now surface integrals over the same surface. The surface is arbitrary, so the integrands are equal:

> **RESULT 2.1: FARADAY'S LAW, DIFFERENTIAL FORM**
>
> $$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \tag{7.25}$$
>
> When $\mathbf{B}$ is constant, this gives $\nabla \times \mathbf{E} = \mathbf{0}$, the electrostatic rule.

4. The universal flux rule

All three experiments give the same formula, whatever changes the flux:

> **RESULT 2.2: UNIVERSAL FLUX RULE**
>
> $$\mathcal{E} = -\frac{d\Phi}{dt} \tag{7.26}$$
>
> Whenever the flux through a loop changes, for whatever reason, an emf $-d\Phi/dt$ appears in the loop.

The book objects to calling Eq. 7.26 "Faraday's law". It combines two mechanisms: a magnetic force on moving charges (Experiment 1) and an induced electric field (Experiments 2 and 3). Keep the two apart when you calculate a particular case.

5. Lenz's law: Example 7.6

A long cylindrical magnet of length L and radius a carries a uniform magnetization $\mathbf{M}$ parallel to its axis. It passes at constant velocity through a ring of slightly larger diameter (Fig. 7.22).

- Before the magnet reaches the ring, the flux through it is zero.
- As the leading end enters, the field of the magnet ($\mathbf{B} = \mu_0 \mathbf{M}$ inside a long magnet) gives a flux that rises to $\mu_0 M \pi a^2$.
- While the magnet is fully inside the ring the flux is constant, so there is no emf.
- As the trailing end leaves, the flux falls back to zero.

So Φ rises, plateaus and falls (Fig. 7.23a), and $\mathcal{E} = -d\Phi/dt$ has two spikes of opposite sign (Fig. 7.23b). The first spike is negative: the first current pulse flows clockwise, and the second counterclockwise, as seen from the left (Lenz's law gives the sense for each spike).

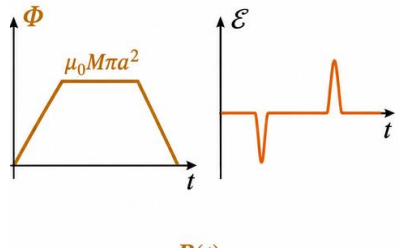


Fig. 7.23 (book): flux and emf for a magnet passing a ring (Example 7.6).

Footnote 14: Lenz's law applies to motional emf too, but there the Lorentz force is usually simpler for the direction.

6. Lenz's law: Example 7.7, the jumping ring

A solenoid with an iron core carries a current, and a metal ring rests on top of it (Fig. 7.24). When the current is switched on, the ring flies off.

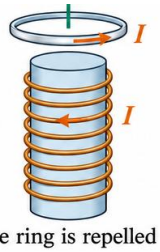


Fig. 7.24 (book): jumping ring on a solenoid with an iron core (Example 7.7).

1. Before the current is switched on, the flux through the ring is zero.
2. After switching on, a flux appears through the ring, pointing up (in the diagram).

3. By Lenz's law, the current induced in the ring produces a field opposing this new flux, so the ring current is opposite to the solenoid current.
4. Opposite currents repel, so the ring is pushed away. This is the jumping ring.

Footnote 15 refers to further studies of the jumping ring and the floating ring.

7.2.2 The induced electric field and the quasistatic approximation

Source: book-sources/m2-sec2-induced.html, Griffiths pp. 320–324. Equations 7.27–7.30; Examples 7.8, 7.9 and 7.10 in full. Problems not included.

Prerequisites

- Faraday's law, Eqs. 7.24 and 7.25.
- Gauss's law and the Biot–Savart law (magnetostatics).

1. Equations for a pure Faraday field

Let the electric field be due only to a changing magnetic field, with no charge ($\rho = 0$). Then

$$\nabla \cdot \mathbf{E} = 0$$
$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

Compare the magnetostatic equations $\nabla \cdot \mathbf{B} = 0$ and $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$. The two systems have the same form, with $\mu_0 \mathbf{J}$ replaced by $-\partial \mathbf{B}/\partial t$. So the induced field can be found by the same methods that give a magnetic field from a current.

2. The Biot–Savart analogue

For a current distribution, the Biot–Savart integral gives $\mathbf{B}$. By the same mathematics, the induced field is

$$\mathbf{E} = -\frac{1}{4\pi} \int \frac{(\partial \mathbf{B}/\partial t) \times \hat{\mathbf{r}}}{r^2} \, d\tau \qquad (7.27)$$

This is correct but rarely the easiest route. When the symmetry allows, use the integral form on an Ampèrian loop. If $\mathbf{E}$ is tangent to the loop and has the same size all round it, the line integral is simply E times the length of the loop, and Faraday's law gives:

RESULT 2.3: FARADAY'S LAW FOR SYMMETRIC CASES

$$\oint \mathbf{E} \cdot d\mathbf{l} = -\frac{d\Phi}{dt} \qquad (7.28)$$

The rate of change of flux through the Ampèrian loop takes the place that $\mu_0 I_{\text{enc}}$ takes in Ampère's law.

PLUS TWO RECALL: EMF ROUND A CIRCULAR LOOP

Recall. From Class 12: the emf round a loop is $-d\Phi/dt$ (Eq. 7.26).

Re-derive.

1. Take a circle of radius s, with $\mathbf{E}$ tangent to it and of the same size all round. The line integral is $\oint \mathbf{E} \cdot d\mathbf{l} = E \cdot 2\pi s$.
2. If $\mathbf{B}$ is uniform over the circle, the flux is $\Phi = \pi s^2 B(t)$.
3. Equate $E \cdot 2\pi s = -d\Phi/dt = -\pi s^2 \, dB/dt$, which gives $E = -\frac{s}{2}\frac{dB}{dt}$.

Saved. Eq. 7.28 applied to a circle.

3. Example 7.8: uniform field over a disk

A uniform field $\mathbf{B}(t)$, pointing straight up, fills a circular region (Fig. 7.25). Find the induced field at radius s.

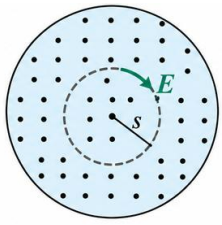


Fig. 7.25 (book): uniform field over a disk, with an Amperian loop of radius s (Example 7.8).

1. By symmetry $\mathbf{E}$ points in the azimuthal direction and has the same size at every point

on a circle of radius s centred on the axis.

2. Apply Eq. 7.28 to that circle: $E(2\pi s) = -\frac{d}{dt}\left(\pi s^2 B(t)\right) = -\pi s^2 \frac{dB}{dt}$.
3. Solve: $\mathbf{E} = -\frac{s}{2}\frac{dB}{dt}\hat{\boldsymbol{\phi}}$.

If $\mathbf{B}$ increases, $\mathbf{E}$ runs clockwise as seen from above.

4. Example 7.9: the wheel

A line charge λ is glued on the rim of a wheel of radius b. The spokes are non-conducting, so the wheel can turn freely. In the central region, out to radius a, there is a field $\mathbf{B}_0$ pointing up, which is switched off (Fig. 7.26).

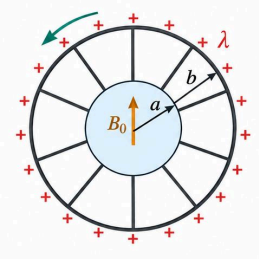


Fig. 7.26 (book): wheel with a line charge on its rim (Example 7.9).

1. Faraday's law on a circle of radius b: $E(2\pi b) = -\pi a^2 \frac{dB}{dt}$, so $\mathbf{E} = -\frac{a^2}{2b}\frac{dB}{dt}\hat{\boldsymbol{\phi}}$ at the rim.
2. Force on a length $d\mathbf{l}$ of rim: $\lambda E\, dl$, with a lever arm b. Torque $= b\lambda E\, dl$.
3. Total torque: $N = b\lambda\left(-\frac{a^2}{2b}\frac{dB}{dt}\right)\oint dl = -b\lambda\pi a^2\frac{dB}{dt}$, using $\oint dl = 2\pi b$.
4. Lenz's law gives the sense of rotation: counterclockwise, as seen from above, so that the wheel's field tends to restore the upward flux.
5. Angular momentum given to the wheel: $\int N\, dt = \lambda\pi a^2 b B_0$. It does not depend on how fast the field is switched off.

The rotation is driven by the electric field. The magnetic field is zero where the charge sits, so the magnetic force on the charge is zero.

5. The quasistatic approximation (Example 7.10)

The field in a changing situation is found from Faraday's law, but the magnetic field on the right-hand side is usually found from magnetostatics. This is accurate when the field changes slowly. That regime is called **quasistatic**. Electromagnetic waves and radiation need the full theory.

Example 7.10: an infinite straight wire carries $I(t)$. Its quasistatic field is $\mathbf{B} = \frac{\mu_0 I}{2\pi s}\hat{\boldsymbol{\phi}}$. To find $\mathbf{E}$, apply Faraday's law to the rectangular Ampèrian loop of Fig. 7.27:

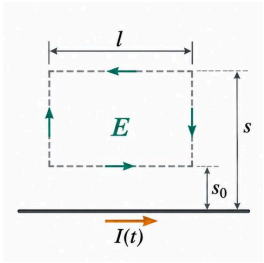


Fig. 7.27 (book): Amperian loop beside a long wire carrying I(t) (Example 7.10).

1. Only the sides parallel to the wire contribute to $\oint \mathbf{E}\cdot d\mathbf{l}$, because $\mathbf{E}$ runs parallel to the axis.
2. The line integral over the loop is

$$E(s_0)l - E(s)l = -\frac{d}{dt}\int \mathbf{B}\cdot d\mathbf{a} = -\frac{\mu_0 l}{2\pi}\frac{dI}{dt}\ln\frac{s}{s_0}$$

3. Hence $\mathbf{E}(s) = \left[\frac{\mu_0}{2\pi}\frac{dI}{dt}\ln s + K\right]\hat{\mathbf{z}}$, where K is a constant (independent of s, but it may depend on t).

$$\mathbf{E}(s) = \left[\frac{\mu_0}{2\pi}\frac{dI}{dt}\ln s + K\right]\hat{\mathbf{z}} \qquad (7.29)$$

The constant K depends on the whole history of $I(t)$. The result grows without limit as $s \to \infty$, which is impossible. The error lies in using the quasistatic approximation far from the wire: electromagnetic news travels at the speed of light, so at large distance $\mathbf{B}$ depends on the current at an earlier time. The approximation holds only for

$$s \ll c\tau \qquad (7.30)$$

where τ is the time for I to change substantially. Eq. 7.29 does not apply at extremely large s.

Footnote 18: the example is artificial, because an infinite wire with the same current at every point at a given instant is unlike a real transmission line. Footnote 19 refers to real coaxial cables.

7.2.3 Inductance

Source: book-sources/m2-sec2-inductance.html, Griffiths pp. 325–330. Equations 7.31–7.38; Examples 7.11, 7.12 and 7.13 in full. Problems not included.

Prerequisites

- Biot–Savart law; Ampère's law; the field of a long solenoid $B = \mu_0 n I$ (Plus Two).
- The vector potential $\mathbf{A}$, with $\nabla \times \mathbf{A} = \mathbf{B}$ and $\mathbf{A} = \frac{\mu_0}{4\pi}\int \frac{\mathbf{J}}{r}\,d\tau$ (Eq. 5.66).
- Faraday's law (Section 7.2.1); Stokes' theorem.

1. Mutual inductance

Take two loops of wire at rest (Fig. 7.30). A steady current I_1 in loop 1 makes a magnetic field $\mathbf{B}_1$. Some of its field lines pass through loop 2. Let Φ_2 be the flux of $\mathbf{B}_1$ through loop 2.

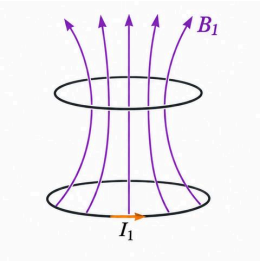


Fig. 7.30 (book): loop 1 carrying I1; its field threads loop 2.

PLUS TWO RECALL: MUTUAL INDUCTANCE

Recall. From Class 12: a changing current in one coil induces an emf in a second coil, through the mutual inductance M.

Re-derive.

1. Biot–Savart: $\mathbf{B}_1 = \frac{\mu_0}{4\pi} I_1 \oint \frac{d\mathbf{l}_1 \times \hat{\mathbf{r}}}{r^2}$.
2. Every factor except I_1 depends only on geometry, so $\mathbf{B}_1 \propto I_1$.
3. Then $\Phi_2 = \int \mathbf{B}_1 \cdot d\mathbf{a}_2 \propto I_1$. Write the constant of proportionality as M_{21}.

Saved. $\Phi_2 = M_{21} I_1$ (Eq. 7.31).

The Biot–Savart law shows that $\mathbf{B}_1$ is proportional to I_1. The flux through loop 2 is then also proportional to I_1:

$$\Phi_2 = M_{21} I_1 \tag{7.31}$$

M_{21} is the mutual inductance of the two loops.

2. The Neumann formula

Write the flux through the vector potential, and use Stokes' theorem:

$$\begin{aligned}\Phi_2 &= \int \mathbf{B}_1 \cdot d\mathbf{a}_2 \\ &= \int (\nabla \times \mathbf{A}_1) \cdot d\mathbf{a}_2 \\ &= \oint \mathbf{A}_1 \cdot d\mathbf{l}_2\end{aligned}$$

The vector potential of loop 1 is $\mathbf{A}_1 = \frac{\mu_0 I_1}{4\pi}\oint \frac{d\mathbf{l}_1}{r}$ (Eq. 5.66). Substituting,

$$\Phi_2 = \frac{\mu_0 I_1}{4\pi}\oint\left(\oint \frac{d\mathbf{l}_1}{r}\right)\cdot d\mathbf{l}_2$$

Comparing with Eq. 7.31 gives the Neumann formula:

RESULT 2.4: NEUMANN FORMULA

$$M_{21} = \frac{\mu_0}{4\pi}\oint\oint \frac{d\mathbf{l}_1 \cdot d\mathbf{l}_2}{r} \tag{7.32}$$

The double integral runs once round each loop. It depends only on the geometry, so M_{21} is purely geometric. Swapping the roles of the loops leaves the integral unchanged, so $M_{21} = M_{12}$ (Eq. 7.33). From now on write M.

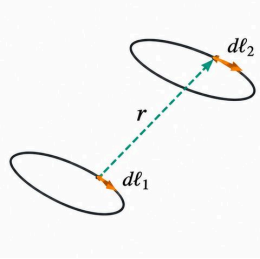


Fig. 7.31 (book): line elements dl1 and dl2 and the vector r (Neumann formula).

The Neumann formula is rarely used for calculation. It shows that the mutual inductance is fixed by the shapes and positions, and that it is symmetric. Swapping the loops gives the same flux per ampere.

3. Example 7.11: a short solenoid inside a long one

A short solenoid (length l, radius a, n_1 turns per unit length) lies on the axis of a very long solenoid (radius b, n_2 turns per unit length). Find M.

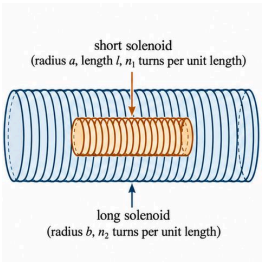


Fig. 7.32 (book): short solenoid inside a long solenoid (Example 7.11).

1. The field of the short solenoid is complicated, and it puts a different flux through each turn of the long solenoid. Computing directly is a miserable task.
2. Use the symmetry $M_{21} = M_{12}$. Run the current I through the long solenoid instead.
3. The field inside a long solenoid is $B = \mu_0 n_2 I$, uniform across the cross-section.
4. The flux through one turn of the short solenoid is $B\pi a^2 = \mu_0 n_2 I \pi a^2$. The short solenoid has $n_1 l$ turns in all.
5. Total flux: $\Phi = \mu_0 \pi a^2 n_1 n_2 I l$.
6. So $M = \mu_0 \pi a^2 n_1 n_2 l$.

If the current in loop 1 changes, Faraday's law gives an emf in loop 2:

$$\mathcal{E}_2 = -\frac{d\Phi_2}{dt} = -M\frac{dI_1}{dt} \tag{7.34}$$

This holds in the quasistatic regime. No wires join the loops; the emf is induced across the gap.

4. Self-inductance

A changing current also induces an emf in its own loop. The field, and therefore the flux through the loop, is proportional to the current:

> **PLUS TWO RECALL: FIELD OF A SOLENOID AND OF A TOROID**
>
> **Recall.** From Class 12: $B = \mu_0 n I$ inside a long solenoid of n turns per unit length.
>
> **Re-derive (toroid, used in Example 7.12).**
>
> 1. Choose a circle of radius s inside the toroid, centred on its axis. By symmetry $\mathbf{B}$ is tangent to the circle and constant on it.
> 2. Ampère's law: $B(2\pi s) = \mu_0 I_{\text{enc}} = \mu_0 N I$, because the circle links all N turns.
> 3. So $B = \frac{\mu_0 N I}{2\pi s}$ inside the toroid, and $B = 0$ outside.
>
> **Saved.** $B = \mu_0 N I / 2\pi s$.

$$\Phi = LI \tag{7.35}$$

L is the self-inductance (or simply the inductance) of the loop. It depends on the geometry and is positive. The emf is

> **RESULT 2.5: SELF-INDUCTANCE EMF**
>
> $$\mathcal{E} = -L\frac{dI}{dt} \tag{7.36}$$
>
> The unit is the henry, H = V·s/A. The minus sign makes the emf a back emf: it opposes any change in current.

5. Example 7.12: toroid

A toroid has a rectangular cross-section (inner radius a, outer radius b, height h) and N turns. Find L.

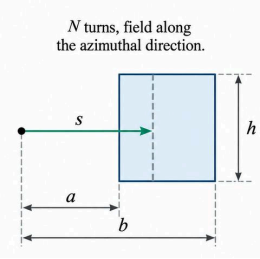


Fig. 7.34 (book): toroid cross-section (Example 7.12).

1. From the box above, $B = \frac{\mu_0 N I}{2\pi s}$.
2. Flux through one turn:
$$\int \mathbf{B}\cdot d\mathbf{a} = \int_a^b \frac{\mu_0 N I}{2\pi s}\, h\, ds = \frac{\mu_0 N I h}{2\pi} \ln\frac{b}{a}$$
.
3. Total flux through all N turns: N times that, $\Phi = \frac{\mu_0 N^2 I h}{2\pi}\ln\frac{b}{a}$.
4. Using $\Phi = LI$ (Eq. 7.35): $L = \frac{\mu_0 N^2 h}{2\pi}\ln\frac{b}{a}$ (Eq. 7.37).

Inductance plays for currents the role that mass plays for velocities. A large L makes the current hard to change. Lenz's law, through the minus sign in Eq. 7.36, makes the back emf oppose any change.

6. Example 7.13: the RL circuit

Example 7.13 has two cases. Cutting a current suddenly makes dI/dt huge, and the back emf produces a spark. Connecting a battery to a resistor and an inductor makes the current grow smoothly, because induction opposes the sudden increase (Fig. 7.35).

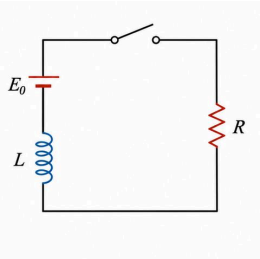


Fig. 7.35 (book): battery, switch, inductor and resistor (Example 7.13).

PLUS TWO RECALL: GROWTH OF CURRENT IN AN LR CIRCUIT

Recall. From Class 12: in an LR circuit the current grows exponentially, with time constant L/R.

Re-derive.

1. Loop rule round the circuit: battery emf $\mathcal{E}_0$, minus the back emf $L\,dI/dt$, equals the drop IR.
2. So $\mathcal{E}_0 - L\frac{dI}{dt} = IR$. This is a first-order equation.
3. Rearrange: $\frac{dI}{dt} = \frac{\mathcal{E}_0 - IR}{L}$. Let $I_\infty = \mathcal{E}_0/R$. Then $\frac{dI}{I - I_\infty} = -\frac{R}{L}dt$.
4. Integrate from $I = 0$ at $t = 0$: $\ln\frac{I_\infty - I}{I_\infty} = -\frac{R}{L}t$, so $I = I_\infty(1 - e^{-Rt/L})$.

Saved. Eq. 7.38 below.

RESULT 2.6: CURRENT GROWTH IN AN RL CIRCUIT

$$I(t) = \frac{\mathcal{E}_0}{R}\left[1 - e^{-(R/L)t}\right] \qquad (7.38)$$

The time constant is $\tau = L/R$. At $t = \tau$ the current has reached about two-thirds of its final value $\mathcal{E}_0/R$. Without inductance the current would jump to $\mathcal{E}_0/R$ at once.

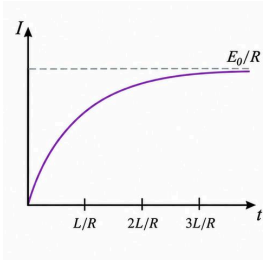


Fig. 7.36 (book): growth of current in an RL circuit.

Footnote 21: the term $-L\,dI/dt$ belongs on the left of the loop equation, because it is part of the emf that drives the voltage across the resistor.

7.2.4 Energy in magnetic fields

Source: book-sources/m2-sec2-energy.html, Griffiths pp. 331–333. Equations 7.39–7.44, the comparison with electrostatic energy, and Example 7.14 in full.

Prerequisites

- $\mathcal{E} = -L\,dI/dt$ (Eq. 7.36); $\Phi = LI$ (Eq. 7.35); vector potential $\mathbf{A}$.
- Ampère's law in differential form, $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$ (magnetostatics).
- Product rule for the divergence of a cross product; the divergence theorem.

1. Work against the back emf

Building up a current takes work. Some of it is lost as heat in resistors, but the rest is stored, and it is returned when the current is switched off. We study the stored part, which is the work done against the back emf.

PLUS TWO RECALL: ENERGY STORED IN AN INDUCTOR

Recall. From Class 12: power is energy per unit time, and the power delivered to a circuit element is $P = VI$.

Re-derive.

1. The work per unit charge against the back emf in one trip round the loop is $-\mathcal{E}$. The minus sign records that it is work done by us, against the emf.
2. Charge passes per unit time I, so the work per unit time is $-\mathcal{E}I$.
3. Use $\mathcal{E} = -L\,dI/dt$: $\frac{dW}{dt} = -\mathcal{E}I = LI\frac{dI}{dt}$.
4. Integrate with respect to time as the current rises from 0 to I: $\int dW = \int LI\,dI$, so $W = \frac{1}{2}LI^2$.

Saved. $W = \frac{1}{2}LI^2$ (Eq. 7.39).

The work per unit time is $dW/dt = -\mathcal{E}I = LI\,dI/dt$. Integrate from zero to the final current I:

RESULT 2.7: ENERGY OF A CIRCUIT

$$W = \frac{1}{2}LI^2 \tag{7.39}$$

This depends only on the geometry (through L) and the final current. It does not depend on how long the process takes.

2. In terms of the vector potential

The flux through the loop is LI (Eq. 7.35). In terms of $\mathbf{A}$ it is $\Phi = \oint \mathbf{A}\cdot d\mathbf{l}$, so $LI = \oint \mathbf{A}\cdot d\mathbf{l}$. Therefore

$$\begin{aligned} W &= \frac{1}{2} I \oint \mathbf{A}\cdot d\mathbf{l} \\ &= \frac{1}{2}\oint (\mathbf{A}\cdot I)\, d\mathbf{l} \end{aligned} \tag{7.40}$$

Generalise to a volume current, replacing $I\,d\mathbf{l}$ by $\mathbf{J}\,d\tau$:

$$W = \frac{1}{2}\int (\mathbf{A}\cdot\mathbf{J})\, d\tau \tag{7.41}$$

3. In terms of the magnetic field

Use Ampère's law, $\mathbf{J} = \nabla\times\mathbf{B}/\mu_0$, to remove $\mathbf{J}$:

$$W = \frac{1}{2\mu_0}\int \mathbf{A}\cdot(\nabla\times\mathbf{B})\, d\tau \tag{7.42}$$

Integrate by parts, using the product rule $\nabla\cdot(\mathbf{A}\times\mathbf{B}) = \mathbf{B}\cdot(\nabla\times\mathbf{A}) - \mathbf{A}\cdot(\nabla\times\mathbf{B})$. Since $\mathbf{B}\cdot(\nabla\times\mathbf{A}) = \mathbf{B}\cdot\mathbf{B}$:

$$W = \frac{1}{2\mu_0}\left[\int_{\mathcal{V}} B^2\, d\tau - \oint_{\mathcal{S}} (\mathbf{A}\times\mathbf{B})\cdot d\mathbf{a}\right] \tag{7.43}$$

Here $\mathcal{S}$ bounds the volume $\mathcal{V}$. The volume must include the current. Any larger volume gives the same W, because $\mathbf{J} = \mathbf{0}$ outside the current, and the surface term shrinks as the surface moves away (both $\mathbf{A}$ and $\mathbf{B}$ fall off). Choose the volume to be all space, where the surface term vanishes:

RESULT 2.8: ENERGY IN THE MAGNETIC FIELD

$$W = \frac{1}{2\mu_0}\int_{\text{all space}} B^2\, d\tau \tag{7.44}$$

The energy density is $B^2/2\mu_0$, which is the usual description of energy "stored in the field".

> Eq. 7.41 instead puts it in the current, at $\frac{1}{2}\mathbf{A}\cdot\mathbf{J}$ per unit volume. Both are bookkeeping; the total W is the physical quantity.

4. Where the energy comes from

Magnetic fields do no work on charges. But building a field requires $\partial\mathbf{B}/\partial t \neq 0$, which induces an electric field, and that field does work while $\mathbf{B}$ grows. There is no $\mathbf{E}$ at the start and none at the end, but there is one in between. Compare the electrostatic energy $W_{\text{elec}} = \frac{\epsilon_0}{2}\int E^2\,d\tau$ (Eq. 2.45). Footnote 22 gives an independent check of Eq. 7.44.

5. Example 7.14: coaxial cable

A long coaxial cable carries current I down the inner cylinder (radius a) and back up the outer one (radius b) (Fig. 7.40). The field lies between the cylinders, $\mathbf{B} = \frac{\mu_0 I}{2\pi s}\hat{\boldsymbol{\phi}}$, and is zero outside.

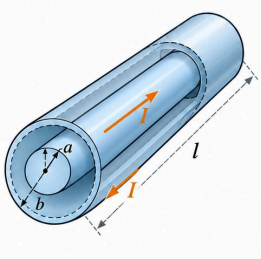


Fig. 7.40 (book): long coaxial cable carrying current I (Example 7.14).

1. Energy per unit volume: $\frac{1}{2\mu_0}\left(\frac{\mu_0 I}{2\pi s}\right)^2 = \frac{\mu_0 I^2}{8\pi^2 s^2}$.
2. A cylindrical shell of length l, radius s and thickness ds has volume $2\pi l s\,ds$. Its energy is $\frac{\mu_0 I^2}{8\pi^2 s^2}\cdot 2\pi l s\,ds = \frac{\mu_0 I^2 l}{4\pi}\frac{ds}{s}$.
3. Integrate from a to b: $W = \frac{\mu_0 I^2 l}{4\pi}\ln\frac{b}{a}$.
4. Compare with $W = \frac{1}{2}LI^2$: $L = \frac{\mu_0 l}{2\pi}\ln\frac{b}{a}$.

This route to L is useful when the current spreads over a surface or volume, because then different parts of the current enclose different fluxes and the flux is hard to find directly. Footnote 23 notes that the rectangular toroid of Example 7.12 is a short coaxial cable turned on its side (compare Eq. 7.37).

7.3.1 Electrodynamics before Maxwell

Source: book-sources/m2-sec3-ampere.html, Griffiths pp. 335–336. Equation 7.45 and the capacitor argument (Fig. 7.43) in full.

Prerequisites

- $\nabla \cdot (\nabla \times \mathbf{A}) \equiv 0$ for any vector field $\mathbf{A}$.
- Continuity equation $\nabla \cdot \mathbf{J} = -\partial \rho / \partial t$ (Eq. 5.29 in the book).
- Ampère's circuital law in integral form (Plus Two).

1. The four laws before Maxwell

Electromagnetic theory before Maxwell consisted of four laws, one for the divergence and one for the curl of each field:

- (i) $\nabla \cdot \mathbf{E} = \rho / \epsilon_0$ (Gauss's law);
- (ii) $\nabla \cdot \mathbf{B} = 0$ (no name);
- (iii) $\nabla \times \mathbf{E} = -\partial \mathbf{B} / \partial t$ (Faraday's law);
- (iv) $\nabla \times \mathbf{B} = \mu_0 \mathbf{J}$ (Ampère's law).

PLUS TWO RECALL: AMPÈRE'S CIRCUITAL LAW

Recall. From Class 12: $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}}$, where I_{enc} is the current through the surface bounded by the loop.

Re-derive (field of a long straight wire).

1. Take a circle of radius s about a long straight wire carrying current I. By symmetry $\mathbf{B}$ is tangent to the circle and has the same size all round.
2. $\oint \mathbf{B} \cdot d\mathbf{l} = B \cdot 2\pi s$, and the enclosed current is I.
3. So $B \cdot 2\pi s = \mu_0 I$, which gives $B = \frac{\mu_0 I}{2\pi s}$.

Saved. $\mathbf{B} = \frac{\mu_0 I}{2\pi s} \hat{\boldsymbol{\phi}}$, the field used in Section 7.2.2.

2. The inconsistency

Take the divergence of (iii). The left side is $\nabla \cdot (\nabla \times \mathbf{E}) = 0$. The right side is $-\partial(\nabla \cdot \mathbf{B})/\partial t$, which is zero by (ii). So (iii) is consistent with (ii).

Now take the divergence of (iv). Again the left side is zero, because $\nabla \cdot (\nabla \times \mathbf{B}) \equiv 0$:

$$\nabla \cdot (\nabla \times \mathbf{B}) = \mu_0 (\nabla \cdot \mathbf{J}) \qquad (7.45)$$

The right side is zero only if $\nabla \cdot \mathbf{J} = 0$, which holds for steady currents. For a charge that builds up somewhere, $\nabla \cdot \mathbf{J} = -\partial \rho / \partial t \neq 0$. So (iv) contradicts the identity, and Ampère's law cannot be right beyond magnetostatics.

3. The capacitor test (Fig. 7.43)

Charge a capacitor through a wire. Apply Ampère's law to a loop round the wire. The loop bounds more than one surface:

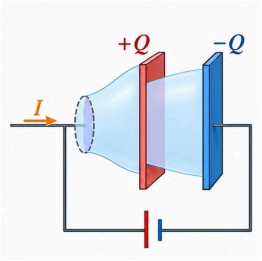


Fig. 7.43 (book): capacitor being charged; Amperian loop with flat and balloon surfaces.

- The plane surface spanning the loop is pierced by the wire, so $I_{\text{enc}} = I$.
- A balloon-shaped surface that passes between the capacitor plates is not pierced by any conduction current, so $I_{\text{enc}} = 0$.

Both surfaces have the same boundary, so the law gives two answers for one loop. "The current enclosed by the loop" is therefore ill-defined when currents are not steady. In magnetostatics this never arises, because charge does not accumulate on capacitor plates.

4. Where the flaw lies

Ampère's law was derived from the Biot–Savart law, so there was no reason to expect it to hold outside magnetostatics. In Maxwell's time there was no experimental evidence against it, so the flaw was purely theoretical. Section 7.3.2 gives Maxwell's repair.

7.3.2 How Maxwell fixed Ampère's law

Source: book-sources/m2-sec3-displacement.html, Griffiths pp. 336–339.

Prerequisites

- Continuity equation; Gauss's law; parallel-plate capacitor $E = \sigma/\epsilon_0$.

1. The correction

$$\nabla \cdot \mathbf{J} = -\frac{\partial \rho}{\partial t} = -\nabla \cdot \left(\epsilon_0 \frac{\partial \mathbf{E}}{\partial t}\right)$$

RESULT 3.1: AMPÈRE–MAXWELL LAW

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \tag{7.46}$$

No change in magnetostatics. Its effect on waves is treated in Chapter 9. Maxwell's ether-based motive is discredited (footnote 24); the charge-conservation argument is the reason that counts.

2. Displacement current

RESULT 3.2

$$\mathbf{J}_d \equiv \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \tag{7.47}$$

$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enc}} + \mu_0 \epsilon_0 \int \frac{\partial \mathbf{E}}{\partial t} \cdot d\mathbf{a} \tag{7.48}$$

$\mathbf{J}_d$ is not a flow of charge; it is added to $\mathbf{J}$. Footnote 25 comments on the word "induce".

PLUS TWO RECALL: PARALLEL-PLATE CAPACITOR

Recall. From Class 12: $E = \sigma/\epsilon_0$ between the plates; $Q = CV$.

Re-derive.

1. $E = Q/\epsilon_0 A$ between the plates.
2. $\partial E/\partial t = \frac{1}{\epsilon_0 A}\frac{dQ}{dt} = \frac{I}{\epsilon_0 A}$.

Saved. $\epsilon_0\, \partial E/\partial t = I/A$ between the plates.

3. Capacitor check

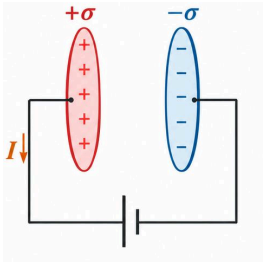


Fig. 7.46 (book): capacitor plates joined by thin wires carrying current I.

- Flat surface (cuts the wire, outside the gap): $E = 0$, $I_{\text{enc}} = I$, displacement term zero.
- Balloon surface (between the plates, not pierced by the wire): $I_{\text{enc}} = 0$; $\int (\partial \mathbf{E}/\partial t) \cdot d\mathbf{a} = I/\epsilon_0$; displacement term $= \mu_0 I$.

Both give $\mu_0 I$.

4. Example 7.15: concentric shells

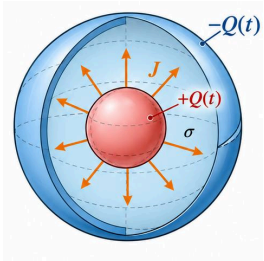


Fig. 7.44 (book): concentric shells with ohmic current between them (Example 7.15).

- Conduction: $\mathbf{J} = \sigma \mathbf{E} = \frac{\sigma Q}{4\pi\epsilon_0 r^2}\hat{\mathbf{r}}$; $I = -\dot{Q} = \sigma Q/\epsilon_0$.
- $\nabla \cdot \mathbf{B} = 0$ with spherical symmetry gives $\mathbf{B} = \mathbf{0}$ (no radial field possible).
- Displacement current $\mathbf{J}_d = \frac{\dot{Q}}{4\pi r^2}\hat{\mathbf{r}} = -\frac{\sigma Q}{4\pi\epsilon_0 r^2}\hat{\mathbf{r}}$ cancels $\mathbf{J}$; so $\nabla \times \mathbf{B} = \mathbf{0}$ and $\mathbf{B} = \mathbf{0}$.

7.3.3 Maxwell's equations

Source: book-sources/m2-sec3-maxwell.html, Griffiths pp. 340–341. Equations 7.49–7.52 in full.

Prerequisites

- The four laws (Section 7.3.1); the Ampère–Maxwell law (Eq. 7.46).
- Gauss's law and the divergence theorem (Plus Two and Section 7.1).

PLUS TWO RECALL: GAUSS'S LAW

Recall. From Class 12: the total flux of $\mathbf{E}$ out of a closed surface is $Q_{\text{enc}}/\epsilon_0$.

Re-derive (differential form).

1. Divergence theorem: $\oint \mathbf{E}\cdot d\mathbf{a} = \int (\nabla\cdot\mathbf{E})\, d\tau$.
2. Enclosed charge: $Q_{\text{enc}} = \int \rho\, d\tau$.
3. Gauss's law holds for every volume, so $\nabla\cdot\mathbf{E} = \rho/\epsilon_0$.

Saved. Maxwell's equation (i), Eq. 7.49.

1. The complete set

Adding the corrected Ampère law (Eq. 7.46) to Gauss's law, the absence of magnetic monopoles, and Faraday's law gives the full set:

RESULT 3.3: MAXWELL'S EQUATIONS (WITH SOURCES)

$$\begin{aligned} \nabla\cdot\mathbf{E} &= \frac{\rho}{\epsilon_0}, \\ \nabla\cdot\mathbf{B} &= 0, \\ \nabla\times\mathbf{E} &= -\frac{\partial\mathbf{B}}{\partial t}, \\ \nabla\times\mathbf{B} &= \mu_0\mathbf{J} + \mu_0\epsilon_0\frac{\partial\mathbf{E}}{\partial t} \end{aligned} \tag{7.49}$$

With the Lorentz force law $\mathbf{F} = q(\mathbf{E}+\mathbf{v}\times\mathbf{B})$ (Eq. 7.50), these contain all of classical electrodynamics apart from the properties of matter. The continuity equation $\nabla\cdot\mathbf{J} = -\partial\rho/\partial t$ (Eq. 7.51) follows from them: take the divergence of the last equation and use the first.

2. The same set, sources on the right

Writing the fields on the left and the sources on the right (Eq. 7.52):

$$\begin{aligned} \nabla\cdot\mathbf{E} &= \frac{\rho}{\epsilon_0}, & \nabla\cdot\mathbf{B} &= 0, \\ \nabla\times\mathbf{E} + \frac{\partial\mathbf{B}}{\partial t} &= \mathbf{0}, & \nabla\times\mathbf{B} - \mu_0\epsilon_0\frac{\partial\mathbf{E}}{\partial t} &= \mu_0\mathbf{J} \end{aligned} \tag{7.52}$$

This layout emphasises that every field has a source: $\mathbf{E}$ comes from charge and from a changing $\mathbf{B}$; $\mathbf{B}$ comes from current and from a changing $\mathbf{E}$.

Maxwell's equations are differential equations, so they need boundary conditions. These are usually obvious from the problem, for example fields vanish far from a localised distribution (footnote 29).

Footnote 28: Maxwell himself wrote about twenty equations, including potentials. Heaviside distilled them into the four equations used today.

7.3.5 Maxwell's equations in matter

Source: book-sources/m2-sec3-matter.html, Griffiths pp. 342–344. Equations 7.56–7.68 in full; Problem 7.42 not included.

Prerequisites

- Bound charge $\rho_b = -\nabla\cdot\mathbf{P}$ (Eq. 7.56); bound current $\mathbf{J}_b = \nabla\times\mathbf{M}$ (Eq. 7.57).
- Continuity equation (Section 7.3.3).

1. Bound charges and currents

Inside polarized or magnetized matter there are bound charges and currents. They are not controlled directly, so it is cleaner to write Maxwell's equations in terms of free charges and currents only.

- Electric polarization $\mathbf{P}$ gives bound charge density $\rho_b = -\nabla\cdot\mathbf{P}$ (Eq. 7.56).
- Magnetization $\mathbf{M}$ gives bound current $\mathbf{J}_b = \nabla\times\mathbf{M}$ (Eq. 7.57).

2. Polarization current

When the fields change in time there is a third current: a changing polarization moves bound charge. Take a short tube of polarized material with $\mathbf{P}$ along its axis (Fig. 7.48).

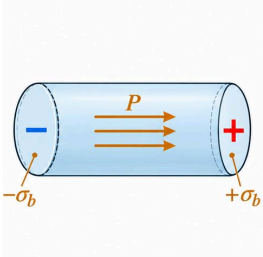


Fig. 7.48 (book): polarized tube with bound charges at its ends.

> **PLUS TWO RECALL: POLARIZATION AND BOUND SURFACE CHARGE**
>
> **Recall.** From Class 12: a dielectric carries bound charge on its surfaces, and its polarization is the dipole moment per unit volume.
>
> **Re-derive.**
>
> 1. The polarization gives a surface charge $\sigma_b = \mathbf{P}\cdot\hat{\mathbf{n}}$ on each end. For $\mathbf{P}$ along the axis, $\sigma_b = +P$ at one end and $-P$ at the other.
> 2. If P increases by $\partial P/\partial t$ per unit time, the charge on each end changes at that rate per unit area.
> 3. So a net current per unit area flows along the tube: $\mathbf{J}_p = \partial\mathbf{P}/\partial t$.
>
> **Saved.** Eq. 7.58.

$$\mathbf{J}_p = \frac{\partial \mathbf{P}}{\partial t} \tag{7.58}$$

This polarization current is different from the bound current $\mathbf{J}_b$. $\mathbf{J}_b$ comes from magnetization (spin and orbital motion of electrons). $\mathbf{J}_p$ is the motion of charge when $\mathbf{P}$ changes.

Check. $\nabla\cdot\mathbf{J}_p = \frac{\partial}{\partial t}(\nabla\cdot\mathbf{P}) = -\frac{\partial \rho_b}{\partial t}$. So $\mathbf{J}_p$ keeps bound charge conserved. A changing magnetization gives no analogous charge accumulation, because $\nabla\cdot(\nabla\times\mathbf{M}) \equiv 0$.

3. Total charge and current

Split the total charge and current into free and bound parts:

$$\rho = \rho_f + \rho_b = \rho_f - \nabla\cdot\mathbf{P} \tag{7.59}$$

$$\begin{aligned}\mathbf{J} &= \mathbf{J}_f + \mathbf{J}_b + \mathbf{J}_p \\ &= \mathbf{J}_f + \nabla\times\mathbf{M} + \frac{\partial \mathbf{P}}{\partial t}\end{aligned} \tag{7.60}$$

4. Gauss's law and Ampère's law in matter

Gauss's law becomes $\nabla\cdot\mathbf{E} = \frac{1}{\epsilon_0}(\rho_f - \nabla\cdot\mathbf{P})$. Rearranging, $\nabla\cdot(\epsilon_0\mathbf{E} + \mathbf{P}) = \rho_f$. Define

$$\mathbf{D} \equiv \epsilon_0\mathbf{E} + \mathbf{P} \tag{7.62}$$

so that $\nabla\cdot\mathbf{D} = \rho_f$ (Eq. 7.61). Here $\mathbf{D}$ is the electric displacement.

$$\nabla\cdot\mathbf{D} = \rho_f \tag{7.61}$$

Ampère–Maxwell becomes $\nabla\times\mathbf{B} = \mu_0\left(\mathbf{J}_f + \nabla\times\mathbf{M} + \partial\mathbf{P}/\partial t\right) + \mu_0\epsilon_0\,\partial\mathbf{E}/\partial t$. Rearranging, $\nabla\times\left(\frac{\mathbf{B}}{\mu_0} - \mathbf{M}\right) = \mathbf{J}_f + \partial\mathbf{D}/\partial t$.

Define

$$\mathbf{H} \equiv \frac{1}{\mu_0}\mathbf{B} - \mathbf{M} \tag{7.64}$$

$$\nabla\times\mathbf{H} = \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t} \tag{7.63}$$

Faraday's law and $\nabla\cdot\mathbf{B} = 0$ are unchanged, since neither contains ρ or $\mathbf{J}$.

> **RESULT 3.4: MAXWELL'S EQUATIONS IN TERMS OF FREE CHARGE AND CURRENT**
>
> $$\begin{aligned}\nabla\cdot\mathbf{D} &= \rho_f, \quad \nabla\cdot\mathbf{B} \\ &= 0, \quad \nabla\times\mathbf{E} \\ &= -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla\times\mathbf{H} \\ &= \mathbf{J}_f + \frac{\partial \mathbf{D}}{\partial t}\end{aligned} \tag{7.65}$$
>
> These are a division of the sources into free and bound parts, not a more general theory than Eq. 7.49. Their mixed notation ($\mathbf{E}$ with $\mathbf{D}$, $\mathbf{B}$ with $\mathbf{H}$) needs constitutive relations.

5. Linear media

For linear media the constitutive relations are

RESULT 3.5: LINEAR MEDIA

$$\mathbf{P} = \epsilon_0 \chi_e \mathbf{E}, \qquad \mathbf{M} = \chi_m \mathbf{H} \tag{7.66}$$

$$\mathbf{D} = \epsilon \mathbf{E}, \qquad \mathbf{H} = \frac{1}{\mu}\mathbf{B} \tag{7.67}$$

with $\epsilon = \epsilon_0(1+\chi_e)$ and $\mu = \mu_0(1+\chi_m)$.

In matter the displacement current is $\mathbf{J}_d = \partial \mathbf{D}/\partial t$ (Eq. 7.68), so the name "displacement current" of Section 7.3.2 is extended to $\partial \mathbf{D}/\partial t$. Problem 7.42 compares conduction and displacement current in sea water (not included).

7.3.6 Boundary conditions

Source: book-sources/m2-sec3-bc.html, Griffiths pp. 345–347. Equations 7.69–7.74 in full.

Prerequisites

- Integral forms of Maxwell's equations in free-charge form (Eq. 7.65).
- Gauss's law for a closed surface; Faraday's law for a loop; divergence theorem.

1. Integral forms

Fields are discontinuous at a boundary between media, and at a surface that carries charge σ or current $\mathbf{K}$. The integral forms of Eq. 7.65 give the jumps:

- (i) $\oint_{\mathcal{S}} \mathbf{D}\cdot d\mathbf{a} = Q_{f,\text{enc}}$ and (ii) $\oint_{\mathcal{S}} \mathbf{B}\cdot d\mathbf{a} = 0$, for any closed surface $\mathcal{S}$;
- (iii) $\oint_{\mathcal{P}} \mathbf{E}\cdot d\mathbf{l} = -\frac{d}{dt}\int_{\mathcal{S}} \mathbf{B}\cdot d\mathbf{a}$ and (iv) $\oint_{\mathcal{P}} \mathbf{H}\cdot d\mathbf{l} = I_{\text{enc}} + \frac{d}{dt}\int_{\mathcal{S}} \mathbf{D}\cdot d\mathbf{a}$, for any surface $\mathcal{S}$ bounded by the loop $\mathcal{P}$.

2. Normal components: the pillbox

PLUS TWO RECALL: FIELD EITHER SIDE OF A SHEET OF CHARGE

Recall. From Class 12: a sheet of surface charge σ gives a jump of σ/ϵ_0 in the normal electric field.

Re-derive.

1. Draw a thin cylinder (pillbox) of cross-section a with one face above and one below the surface.
2. Apply (i). The flux out through the two faces is $D^{\perp}_{\text{above}}a - D^{\perp}_{\text{below}}a$. The curved side vanishes as its height goes to zero.
3. The free charge enclosed is $\sigma_f a$.

Saved. $D^{\perp}_{\text{above}} - D^{\perp}_{\text{below}} = \sigma_f$ (Eq. 7.69).

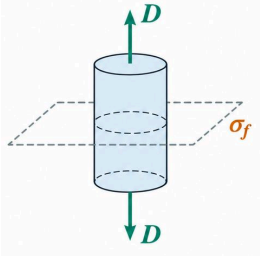


Fig. 7.49 (book): thin pillbox straddling a charged surface.

Applying (i) to a thin pillbox that extends slightly into each medium gives $\mathbf{D}_{\text{above}}\cdot\mathbf{a} - \mathbf{D}_{\text{below}}\cdot\mathbf{a} = \sigma_f d$, where $\mathbf{a}$ points upward and d is the pillbox thickness. The edge and any volume charge contribute nothing in the limit $d \to 0$:

RESULT 3.6: NORMAL COMPONENTS

$$D^{\perp}_{\text{above}} - D^{\perp}_{\text{below}} = \sigma_f \tag{7.69}$$

$$B^{\perp}_{\text{above}} - B^{\perp}_{\text{below}} = 0 \tag{7.70}$$

Eq. 7.70 follows in the same way from (ii).

3. Tangential components: the thin loop

A very thin rectangular loop straddles the surface. Apply (iii): $\mathbf{E}_{\text{above}} \cdot \mathbf{l} - \mathbf{E}_{\text{below}} \cdot \mathbf{l} = -\frac{d}{dt} \int \mathbf{B} \cdot d\mathbf{a}$. As the loop's width goes to zero, the flux through it goes to zero, and the contributions of its short ends also vanish. So

> **RESULT 3.7: TANGENTIAL E IS CONTINUOUS**
>
> $$\mathbf{E}^{\parallel}_{\text{above}} - \mathbf{E}^{\parallel}_{\text{below}} = \mathbf{0} \tag{7.71}$$

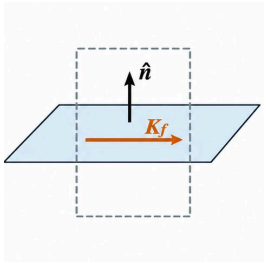


Fig. 7.50 (book): thin Amperian loop straddling a surface current.

For (iv), take $\hat{\mathbf{n}}$ as the unit normal pointing up, so that the loop's normal to the surface is $\hat{\mathbf{n}} \times \mathbf{l}$. The enclosed free current is $I_{f,\text{enc}} = \mathbf{K}_f \cdot (\hat{\mathbf{n}} \times \mathbf{l}) = (\mathbf{K}_f \times \hat{\mathbf{n}}) \cdot \mathbf{l}$. A volume current gives nothing in the thin limit, but a surface current does:

> **RESULT 3.8: TANGENTIAL H JUMPS BY THE SURFACE CURRENT**
>
> $$\mathbf{H}^{\parallel}_{\text{above}} - \mathbf{H}^{\parallel}_{\text{below}} = \mathbf{K}_f \times \hat{\mathbf{n}} \tag{7.72}$$
>
> The parallel components of $\mathbf{H}$ jump by an amount proportional to the free surface current.

4. Linear media

With $\mathbf{D} = \epsilon\mathbf{E}$ and $\mathbf{H} = \mathbf{B}/\mu$, the conditions become (Eq. 7.73):

$$\begin{aligned} \epsilon_{\text{above}} E^{\perp}_{\text{above}} - \epsilon_{\text{below}} E^{\perp}_{\text{below}} &= \sigma_f, \\ B^{\perp}_{\text{above}} - B^{\perp}_{\text{below}} &= 0, \\ E^{\parallel}_{\text{above}} - E^{\parallel}_{\text{below}} &= 0, \\ \frac{B^{\parallel}_{\text{above}}}{\mu_{\text{above}}} - \frac{B^{\parallel}_{\text{below}}}{\mu_{\text{below}}} &= \mathbf{K}_f \times \hat{\mathbf{n}} \end{aligned} \tag{7.73}$$

With no free charge or free current at the interface, the right-hand sides vanish (Eq. 7.74). These are the conditions used to derive reflection and refraction of waves (Chapter 9).

Background: magnetic charge (7.3.4) and the rotating magnet (7.4)

Source: book-sources/m2-sec3-background.html, Griffiths pp. 341–350. ***Outside syllabus items 6–8.*** *Kept for completeness; no equations dropped.*

1. Magnetic charge (§7.3.4)

In free space ($\rho = 0$, $\mathbf{J} = \mathbf{0}$) the four equations are symmetric. Replacing $\mathbf{E} \to \mathbf{B}$ and $\mathbf{B} \to -\mu_0\epsilon_0\mathbf{E}$ swaps the two pairs. If magnetic charge density ρ_m and magnetic current $\mathbf{J}_m$ existed, the full set would be (Eq. 7.53):

$$\nabla \cdot \mathbf{E} = \frac{\rho_e}{\epsilon_0}, \qquad \nabla \cdot \mathbf{B} = \mu_0 \rho_m,$$
$$\nabla \times \mathbf{E} = -\mu_0 \mathbf{J}_m - \frac{\partial \mathbf{B}}{\partial t}, \qquad \nabla \times \mathbf{B} = \mu_0 \mathbf{J}_e + \mu_0\epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \tag{7.53}$$

Each charge would be conserved (Eq. 7.54): $\nabla \cdot \mathbf{J}_m = -\partial\rho_m/\partial t$ and $\nabla \cdot \mathbf{J}_e = -\partial\rho_e/\partial t$. The force between magnetic charges is Eq. 7.55, $\mathbf{F} = \frac{\mu_0}{4\pi}\frac{q_{m1}q_{m2}}{r^2}\hat{\mathbf{r}}$.

No magnetic monopole has been found, and Maxwell's equations do not contain one. Magnetic charge would explain why electric charge is quantised (Dirac; Problem 8.21). Footnotes 31–33 give the search literature.

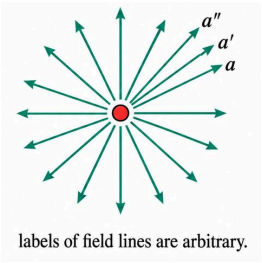


Fig. 7.52 (book): field lines of a point charge; labels are arbitrary.

2. The rotating magnet (§7.4)

Faraday's experiments with a disk and a magnet (Fig. 7.51) show that a magnet rotating about its own symmetry axis does not drag its field round. A magnet rotating about a perpendicular axis does drag its field along. In 1912 Barnett concluded that the Earth's field, as associated with the Earth's rotation about its axis, is not dragged.

In Maxwell's theory the fields are the physical quantities, and field lines are a construct for drawing them. The labelling of field lines has no physical content (Fig. 7.52), so the statement that field lines rotate with their source is a choice of description, not a physical result. Only the magnitude and direction at each point matter.

The Lorentz force uses the charge's velocity relative to a frame, which makes the magnetic case subtle. Footnote 36 corrects footnote 34: moving magnetic field lines do not induce electric fields; time-dependent fields do.